

```
osfy@Shaanu:~/docker-vs-wasm$ $WASI_SDK_PATH/bin/clang --target=wasm32-wasi -O2 -o hello.wasm hello-w.c
osfy@Shaanu:~/docker-vs-wasm$ ls -lh hello.wasm
-rwxrwxr-x 1 osfy osfy 4.4K Jun 10 12:37 hello.wasm
osfy@Shaanu:~/docker-vs-wasm$ /usr/bin/time -f "Execution Time (WASM): %e seconds" wasmtime hello.wasm
Hello from Web Assembly !
Execution Time (WASM): 0.01 seconds
osfy@Shaanu:~/docker-vs-wasm$
```

Figure 3: Wasm binary creation and execution

In contrast, the WebAssembly version, compiled with the WASI SDK and run using Wasmtime, produced a compact .wasm binary of just 4.4KB and an execution time of 0.01 seconds as shown in Figure 3. This demonstrates the enormous efficiency gains that Wasm offers for small, fast-starting, and isolated workloads. The result also reflects Wasm's core design goals — low overhead, secure sandboxing, and cross-platform portability.

Looking forward, WebAssembly is expected to gain even more traction as the ecosystem matures. The addition of features such as multi-threading, garbage collection, and native support for more programming languages will enable developers to use Wasm for increasingly complex applications. With cloud platforms and edge environments

embracing Wasm, we may soon see it integrated seamlessly into CI/CD pipelines and microservice architectures. Meanwhile, Docker will continue to evolve, possibly integrating Wasm runtimes and expanding its support for lightweight execution models. The future lies in hybrid solutions that combine the full-stack capabilities of containers with the speed and portability of Wasm. **END** 🐧

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